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Exercise-Induced Laryngeal Obstruction (EILO) versus Exercise-Induced Asthma in Athletes: A Narrative Review of Differentiation and Management

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Abstract

Background: Exercise-induced laryngeal obstruction (EILO) and exercise-induced bronchoconstriction (EIB), the clinical hallmark of exercise-induced asthma, are the two dominant medical explanations for exertional dyspnoea in athletes. They are physiologically distinct yet routinely confused, and EILO is frequently mislabelled and mistreated as refractory or difficult-to-control asthma.

Aim: To critically compare EILO and exercise-induced asthma in athletes and to synthesise current evidence on their differentiation and clinical management.

Material and methods: Narrative review of peer-reviewed clinical studies, consensus documents and reviews addressing the pathophysiology, epidemiology, diagnosis and treatment of EILO and EIB in athletic and adolescent populations.

Results: EILO is a mechanical, predominantly supraglottic, inspiratory upper-airway collapse that peaks at maximal effort and resolves within minutes, whereas EIB is an inflammatory, expiratory lower-airway obstruction that typically peaks after exercise. Continuous laryngoscopy during exercise is the diagnostic gold standard for EILO, while EIB requires objective bronchoprovocation. Symptoms and questionnaires cannot reliably separate the two conditions, which frequently coexist. EILO responds to non-pharmacological strategies and does not respond to asthma medication or inhaled ipratropium, whereas EIB is managed with

inhaled corticosteroids and beta-2 agonists within anti-doping limits.

Conclusions: Persistent exertional dyspnoea that fails to respond to asthma therapy should prompt objective testing for EILO rather than escalation of asthma treatment. Awareness, objective diagnosis and condition-specific management are essential to avoid medical error and impaired performance.

Keywords: exercise-induced laryngeal obstruction; EILO; exercise-induced bronchoconstriction; exercise-induced asthma; athletes; differential diagnosis; exertional dyspnoea.

1. Introduction

Respiratory complaints are among the most common medical problems reported by athletes, and exertional dyspnoea in particular carries an importance that extends well beyond ordinary breathlessness: in elite sport the boundary between success and failure is frequently drawn by the efficiency of the respiratory system. For several decades asthma and its exercise-related manifestation, exercise-induced bronchoconstriction (EIB), were regarded as almost the sole medical explanation for breathing difficulties provoked by exertion, and clinical work, research and even sports legislation were organised around the assessment and treatment of the lower airways [1]. This historical focus was not misplaced, because asthma and EIB are genuinely common in athletes; it was, however, incomplete. Over the last two decades exercise-induced laryngeal obstruction (EILO) has been established as a distinct and clinically important cause of exertional breathing problems that mimics lower-airway disease but originates in the larynx, at the very top of the airway tree [1].

EILO is defined as a transient, inappropriate adduction of laryngeal structures during exercise in a larynx that is anatomically and functionally normal at rest, producing exertional dyspnoea that is often accompanied by noisy, mainly inspiratory breathing and stridor [1]. The condition affects an estimated five to eight per cent of adolescents in the general population, and considerably more—more than twenty per cent in some cohorts—among elite athletes and other individuals whose everyday life demands very high ventilation [1]. Because the presenting symptoms of EILO overlap so extensively with those of EIB, the two entities are habitually confused by athletes, coaches, support staff and clinicians alike, and undiagnosed EILO is frequently perceived and managed as untreated or under-treated asthma [2]. The practical consequences of this confusion are far from trivial: over-prescription of asthma medication and its attendant side effects, unwarranted applications for therapeutic use exemptions, wasted healthcare resources, and a persistence of disabling symptoms that can curtail sporting careers [1,2].

Physiologically, the two conditions could hardly be more different. Whereas EIB reflects an inflammatory narrowing of the lower airways, EILO is a mechanical event at the level of the larynx, driven by aerodynamic forces that arise when the upper airway becomes an unstable bottleneck during intense effort [3]. This mechanistic divergence explains why the treatments differ so profoundly, and why the anti-inflammatory inhalers that benefit asthma have no rational place in the management of a suction-driven laryngeal collapse [3]. Yet, in day-to-day sports medicine, the two conditions continue to be conflated, and athletes with EILO are routinely subjected to prolonged and ineffective courses of asthma pharmacotherapy before the correct diagnosis is finally considered [2,3].

Compounding the difficulty, exercise-induced asthma is itself a heterogeneous entity, with a prevalence in athletes far higher than in the general population and a phenotype that differs from classical allergic asthma in its inflammatory profile and its response to treatment [4,5]. The breathless athlete therefore presents the clinician with a genuine diagnostic conundrum in

which two common, physiologically opposite, and frequently coexisting conditions must be disentangled before rational management can begin [2,4]. The purpose of this narrative review is to compare EILO and exercise-induced asthma in athletes across their pathophysiology, epidemiology, clinical presentation, diagnosis and treatment, to examine critically why they are so often confused, and to synthesise the current evidence on how they can be differentiated and managed. The emphasis throughout is on the medical aspects of differentiation and on the clinical reasoning that separates a mechanical upper-airway disorder from an inflammatory lower-airway disease.

2. Material and Methods

This work is a narrative review synthesising the current literature on EILO and exercise-induced asthma in athletes, with a deliberate focus on their differential diagnosis and management. The evidence base comprises peer-reviewed original clinical studies, consensus and position statements, systematic and narrative reviews, and illustrative case reports addressing the pathophysiology, epidemiology, diagnosis and treatment of laryngeal and lower-airway causes of exertional dyspnoea in athletic and adolescent populations. Priority was given to studies employing objective diagnostic methods—continuous laryngoscopy during exercise for EILO and standardised bronchoprovocation for EIB—and to work concerning competitive athletes and physically active young people, in whom the diagnostic overlap is most consequential.

Consistent with the aims of a narrative rather than a systematic review, the material was interrogated critically: convergent and divergent findings were compared, methodological limitations were weighed, and the strength of the underlying evidence was appraised. Where the literature disagreed—for example on the prevalence of EILO across differing test protocols, on the diagnostic thresholds used to define laryngeal obstruction, or on the efficacy of individual treatments—the discrepancies are highlighted rather than smoothed over. No new patient data were generated, and the review therefore did not require ethical approval.

3. Terminology and Nomenclature

Precise terminology is a prerequisite for any meaningful comparison of these conditions, because both have suffered from a proliferation of overlapping and inconsistent labels. Laryngeal obstruction provoked by exercise was for many years described as vocal cord dysfunction (VCD) or paradoxical vocal fold motion, terms that focused attention on the glottis and the vocal folds. By international consensus, breathing problems caused by inappropriate laryngeal narrowing in an otherwise normal larynx are now designated inducible laryngeal obstruction (ILO), with EILO reserved for the situation in which exercise is the inducer [1]. This reclassification is more than semantic. Continuous laryngoscopy performed throughout exercise has demonstrated that, in the majority of patients, obstruction begins at the supraglottic level—through inward collapse of the aryepiglottic folds and arytenoid structures—and that adduction of the vocal folds, when it occurs, is usually a secondary phenomenon following the supraglottic event [1]. The older labels VCD and paradoxical vocal fold motion, which imply a purely glottic mechanism, are therefore misleading and have largely been abandoned in favour of the ILO/EILO nomenclature [1,6].

It should nonetheless be recognised that VCD, paradoxical vocal fold motion and ILO/EILO continue to be used interchangeably in parts of the clinical and speech-pathology literature, which complicates comparison across studies and contributes to the low awareness of the condition among non-specialists [6,7]. In broader classifications of dysfunctional breathing, ILO—including its exercise-induced subtype—is grouped with extrathoracic or upper-airway breathing pattern disorders, distinguishing it conceptually from thoracic dysfunctional breathing and from lower-airway disease [6]. A pragmatic phenotypic scheme further separates

classic ILO, lung-disease-associated ILO, exercise-associated ILO and irritant-associated ILO, reflecting the differing triggers and, probably, differing mechanisms of laryngeal closure [7].

The terminology of exercise-related lower-airway obstruction has undergone a parallel refinement. The historical term exercise-induced asthma (EIA) is now discouraged because it wrongly implies that exercise causes asthma, when in reality exertion precipitates or worsens bronchoconstriction in a person with underlying airway disease [8,9]. Contemporary usage distinguishes EIB with asthma (EIBa), denoting exercise-triggered bronchial obstruction in a patient with clinical asthma, from EIB without asthma (EIBwa), denoting the same physiological event in an individual without other features of asthma [8,9,10]. EIB is thus the preferred umbrella description of the acute, transient airway narrowing that occurs during or after exertion, whether or not chronic asthma is present [8,10]. In this review the phrases exercise-induced asthma and EIB are used to denote the lower-airway, inflammatory, bronchoconstrictive process, in deliberate contrast to EILO, the upper-airway, mechanical, laryngeal process; where a distinction between EIBa and EIBwa is material, it is made explicit.

4. Normal Laryngeal Function and the Pathophysiology of EILO

The larynx sits above the trachea as a narrow, complex, valve-like structure that regulates airflow to the lungs, protects them from aspiration, and contributes to phonation and to effective cough [1]. It represents and controls a large proportion of total airway resistance, and its patency is therefore a decisive determinant of ventilatory capacity during exercise. In a healthy response to exertion, the vocal folds and arytenoids move into a relatively fixed, abducted position during both inspiration and expiration, while the epiglottis flattens against the tongue base to stretch the aryepiglottic folds; these coordinated movements enlarge the cross-sectional area of the laryngeal aperture and optimise airflow precisely when ventilatory demand is greatest [1]. The importance of this structure is underlined by the demonstration that translaryngeal airway resistance can be measured reproducibly during maximal exercise and that it is higher in females than in males, a finding consistent with the smaller inner laryngeal dimensions of the female larynx after puberty and with the female predominance of EILO [12].

EILO arises when this normal response fails and the larynx cannot maintain patency against the aerodynamic forces generated by high-volume ventilation. The condition is best understood through fluid dynamics rather than inflammation: as inspiratory airflow velocity increases through the narrowest point of the airway, lateral pressure within the lumen falls in accordance with Bernoulli's principle, generating a medially directed suction that draws the laryngeal tissues inward [3]. Because increasing inspiratory effort raises airflow velocity still further, a self-reinforcing loop is created in which narrowing intensifies suction, which hastens collapse—a paradox whereby the harder the athlete tries to breathe in, the more the airway closes [3]. Susceptibility is thought to reflect a combination of relative anatomical flaccidity of the supraglottic structures and imperfect neuromuscular coordination, such that the posterior cricoarytenoid abductors fail to hold the airway open in perfect synchrony with the surging inspiratory flow [3].

Computational fluid dynamic modelling of the upper airway supports this aerodynamic account and has begun to illuminate why particular breathing manoeuvres are protective. Simulations demonstrate that the narrowest cross-section and the highest airflow velocities occur at the glottic plane, that deep mouth breathing generates the largest negative intraluminal pressures and hence the greatest collapsing forces, and that nasal breathing and controlled techniques such as a quick sniff on inspiration and pursed-lip breathing on expiration are associated with more favourable pressure profiles and a wider laryngeal aperture [11]. These models provide a mechanistic rationale for the breathing strategies used in the conservative treatment of EILO, and they reinforce the conceptual distinction between a suction-driven mechanical failure of the

upper airway and the inflammatory narrowing of the lower airway that characterises EIB [11].

Perhaps the most important recent insight into EILO pathophysiology is that a degree of exercise-induced laryngeal narrowing may be part of the normal spectrum of the athletic airway rather than a categorical disease state. When healthy, well-trained endurance athletes with no self-reported breathing problems were examined with continuous laryngoscopy during maximal exercise, one third displayed supraglottic obstruction of a grade conventionally considered compatible with EILO, and none had isolated glottic obstruction [13]. This finding, unexpected by the investigators, indicates that the laryngeal response to very high airflow is diverse and that visualised obstruction alone, in the absence of compatible symptoms, is insufficient to establish the diagnosis [13]. It also cautions strongly against over-diagnosis of EILO in athletes if diagnostic decisions rest solely on the appearance of the larynx during exercise, and it reframes EILO as a condition defined by the conjunction of laryngeal obstruction and symptoms rather than by laryngeal obstruction alone [13].

Emerging mechanistic work suggests that the supraglottic and glottic subtypes of EILO may arise through partly different mechanisms. Supraglottic collapse is thought to be triggered by the high airflow rates of exercise creating inward and downward forces on the aryepiglottic folds and arytenoids, forces that increase steeply with minute ventilation and are influenced by the individual geometry of the hypopharynx, so that anatomical shape interacts with aerodynamic load to determine susceptibility [3,24]. Glottic adduction, by contrast, has been linked to neurological factors including an irritable or hypersensitive laryngeal closure reflex, although studies of laryngeal sensitivity in EILO have produced conflicting results, some reporting decreased rather than increased sensitivity [24]. The practical significance of this distinction is that it maps onto the therapeutic division between conservative treatment—effective particularly at the glottic level—and surgery, which is reserved for anatomical supraglottic collapse, reinforcing the requirement that the CLE test define not merely the presence but the level and character of the obstruction [1,24].

The causes and risk factors for EILO remain incompletely understood. Asthma, gastro-oesophageal reflux, nasal disease, anatomical characteristics of the upper airway, behavioural-health factors and possible genetic influences have all been proposed as contributors or associations, but none has been proven to be causal, and treatment directed at putative contributors such as reflux has not reliably relieved symptoms [1]. The female predominance that emerges around and after puberty is plausibly linked to the smaller female laryngeal diameter and to the greater translaryngeal resistance measured in women, although this remains a hypothesis requiring further study [1,12]. A psychological component has often been invoked, but a systematic over-representation of psychiatric disease or personality disorder has not been demonstrated in individuals with isolated EILO, and heightened anxiety appears no more common than in control populations; if anything, the fear and panic that can accompany an episode of laryngeal closure are as likely to be a consequence of the frightening sensation of choking as a cause of it [1]. Taken together, the evidence portrays EILO as a dynamic, mechanical, aerodynamically driven disorder of the upper airway whose expression depends on the interaction between very high airflow and individual laryngeal anatomy and control [1,3,11].

The conceptualisation of the larynx as the unexplored bottleneck of the airways has reframed EILO from a curiosity into a mechanistically coherent disorder of upper-airway flow limitation [1]. Work-of-breathing studies indicate that the development of EILO is accompanied by measurable increases in respiratory neural drive and in the effort of breathing, giving objective substance to the common athlete complaint that breathing simply feels harder, and translaryngeal pressure and resistance measurements are beginning to quantify the functional cost of the visualised narrowing [1,12]. A hereditary contribution to supraglottic EILO has been

proposed on the basis of familial clustering, and congenital laryngomalacia has been linked to the later development of exercise-induced laryngeal obstruction, suggesting that pre-existing laxity of the supraglottic structures may predispose to collapse under the extreme airflow of exercise; these observations remain preliminary, and no definite mode of inheritance has been established [1]. Together they portray EILO as a disorder in which relatively fixed anatomical characteristics interact with the dynamic aerodynamic stress of high ventilation to produce intermittent, effort-dependent obstruction [1,3].

5. Pathophysiology of Exercise-Induced Bronchoconstriction and Asthma

Exercise-induced bronchoconstriction is, by contrast, a disorder of the lower airways in which the fundamental event is transient bronchial smooth-muscle contraction and airway-wall oedema provoked by the high ventilation of exercise [4,9]. Two complementary and probably cooperative mechanisms are invoked. According to the osmotic theory, the large volumes of relatively cold, dry air inhaled during hyperpnoea evaporate water from the airway surface liquid, rendering it hyperosmolar; this osmotic stimulus drives water out of epithelial and inflammatory cells and triggers the release of bronchoconstrictor mediators—histamine, cysteinyl leukotrienes and prostaglandins—from mast cells and eosinophils, producing smooth-muscle contraction and mucus secretion [4,9,10]. According to the thermal or vascular theory, cooling of the airway during exercise followed by rapid rewarming afterwards causes reactive hyperaemia and vascular engorgement of the bronchial microcirculation, with airway-wall oedema and consequent narrowing [4,9]. Airway epithelial injury from the repeated dehydration and mechanical stress of high ventilation is increasingly recognised as an amplifying factor, particularly in athletes exposed to noxious inhalants [5,9].

The inflammatory phenotype of the athletic airway differs in important respects from that of classical allergic asthma. Whereas asthmatic bronchoconstriction is typically associated with eosinophilic inflammation that responds to inhaled corticosteroids, the airway inflammation seen in many elite athletes, especially those training in cold air, has a substantial neutrophilic component that does not correlate consistently with bronchial hyper-responsiveness and does not respond as reliably to inhaled steroids [4,14]. Studies of cross-country skiers and swimmers have documented airway epithelial damage, remodelling and mixed inflammatory infiltrates acquired over seasons of training, giving rise to the concept of a distinct sports-asthma phenotype driven by the occupational stress of repeated high-volume ventilation in adverse environments rather than by atopy alone [5,14]. Vascular and endothelial factors may also participate; increased bronchial vascularity and mediators such as vascular endothelial growth factor have been implicated in airway remodelling, and markers of endothelial dysfunction have been studied in the context of asthma and its treatment [5,15].

The environment in which an athlete trains is a decisive modifier of EIB risk and helps to explain the striking variation in prevalence between sports. Endurance and winter-sport athletes inhale enormous volumes of cold, dry air, which favours the osmotic and thermal mechanisms of bronchoconstriction, and cross-country skiers and other cold-air athletes show a high prevalence of airway hyper-responsiveness and asthma that increases with years of competition [5,14]. Swimmers are exposed to chlorine disinfection by-products such as trichloramine, which irritate and injure the airway epithelium; the relationship between chlorinated-pool attendance and the development of asthma and EIB has been extensively debated, with some evidence of harm and other data suggesting a degree of protection from the humid pool environment, so that the pool-chlorine hypothesis remains unresolved and in need of long-term study [5,9,16]. Athletes competing on ice are exposed to nitrogen oxides and ultrafine particulate matter from ice-resurfacing machines, and outdoor endurance athletes to ambient pollutants and aeroallergens, each of which can aggravate airway inflammation and provoke

bronchoconstriction [5,9,16]. These environmental exposures are relevant to the differential diagnosis because the same conditions—cold, dry air in particular—have been reported anecdotally to worsen EILO as well, so that environmental provocation alone does not distinguish upper- from lower-airway disease [1].

At the cellular level, the airway narrowing of EIB is the end result of a cascade initiated by dehydration and cooling of the airway surface. Hyperosmolarity of the airway lining fluid provokes degranulation of mast cells and eosinophils and the release of histamine, cysteinyl leukotrienes and prostaglandins, which contract airway smooth muscle and increase mucus production, while epithelial injury from repeated high-volume ventilation releases damage-associated molecular patterns that activate macrophages and drive the secretion of pro-inflammatory cytokines such as tumour necrosis factor and interleukins-1, -6 and -8 [5,9,10]. Elevated urinary and serum concentrations of Clara cell secretory protein-16 after exercise provide a biochemical signature of this epithelial stress, and neurogenic mechanisms—increased release of substance P and activation of cholinergic pathways, possibly through transient receptor potential channels—further amplify bronchoconstriction and secretion [5,9]. These converging osmotic, inflammatory and neurogenic pathways explain both the reversibility of EIB and its tendency, in athletes chronically exposed to injurious inhalants, to be accompanied by airway remodelling [5,14].

Atopy is a major amplifier of EIB in children and a frequent, though not universal, feature in athletes; up to forty per cent of children with allergic rhinitis experience EIB, and sensitisation to indoor allergens, elevated immunoglobulin E and atopic dermatitis increase the risk, so that comorbid rhinitis and allergy should be sought and treated in the breathless young athlete [5,9]. Nevertheless, a substantial subgroup of athletes—the so-called sports-asthma phenotype—display exercise-induced respiratory symptoms and airway hyper-responsiveness with little atopy and predominantly neutrophilic rather than eosinophilic inflammation, and in these individuals inhaled corticosteroids are often less effective, a therapeutically important distinction from classical allergic asthma [5,14]. This heterogeneity of the lower-airway phenotype is directly relevant to the differential with EILO, because an athlete whose exertional dyspnoea does not respond to inhaled corticosteroids may have non-eosinophilic sports asthma, may have EILO, or may have both, and only objective testing can decide between these possibilities [5,14].

A genetic contribution to EIB has been suggested, with candidate genes including that encoding preprotachykinin (the precursor of substance P), the aquaporin-5 water-channel gene implicated in airway reactivity, and the Clara cell secretory protein gene, although the association between genetic variation and EIB has not been conclusively established [5,9]. Intense and prolonged training may also induce a transient immune downregulation with a shift toward a T-helper-2 response, clinically associated with increased atopy and susceptibility to upper-respiratory infection, both of which are risk factors for the onset and worsening of airway disease in athletes [5]. These observations situate the athletic airway at the intersection of genetic predisposition, immune modulation and environmental exposure, and they help to explain why the sports-asthma phenotype is heterogeneous and does not always conform to the eosinophilic, corticosteroid-responsive pattern of classical asthma [5,14].

6. Epidemiology: EILO and EIB in Athletes and Adolescents

Both conditions are common, but their epidemiology is shaped by the population studied and, critically, by the diagnostic method used. For EILO, prevalence estimates in the general adolescent population cluster around five to eight per cent, rising to more than twenty per cent in groups for whom high-intensity exercise is routine, such as elite athletes and military recruits [1]. These figures are limited by variable diagnostic criteria and by referral bias, and many older

studies lack confirmation by continuous laryngoscopy during exercise; in the most rigorously tested athlete cohorts the reported prevalence has ranged widely depending on the population and the exercise protocol applied [1]. A cross-sectional study of adolescent athletes attending a sports high school, in which randomly selected participants underwent both an EIB test and a continuous laryngoscopy exercise test, estimated the prevalence of EILO at approximately eight per cent and that of EIB at approximately twenty-three per cent, and—importantly—found that self-reported exercise-induced dyspnoea was a weak indicator of both conditions [17]. This dissociation between symptoms and objective findings is a recurring theme and a central argument for objective testing.

In populations selected for exertional respiratory symptoms, the yield of EILO is considerably higher. Among patients referred for continuous laryngoscopy because of exercise-induced breathing problems despite prior pulmonary assessment, EILO was diagnosed in about forty per cent, with the highest prevalence—over sixty per cent—in females under eighteen years of age, but with clinically important rates also in adolescent males and in adults of both sexes, underscoring that EILO is not confined to young female athletes [18]. In a selected group of national- and international-level athletes referred specifically for suspected EILO, the diagnosis was confirmed by continuous laryngoscopy in nearly three quarters, of whom the great majority had moderate to severe supraglottic obstruction and only a small minority had predominant glottic obstruction [2]. The heterogeneity of these prevalence estimates is not a weakness of the literature so much as a reflection of the profound influence of case selection and diagnostic rigour, and it reinforces the point that EILO is under-recognised precisely because it is not sought with the appropriate test [2,17,18].

For EIB the epidemiological picture is at once clearer and larger in scale. Prevalence in the general population is of the order of five to twenty per cent, but in athletes it rises substantially and varies markedly by sport, reaching thirty to seventy per cent in some endurance and winter disciplines [8,19]. A systematic review of EIB among adolescent athletes reported prevalences ranging from about two to sixty-one per cent, a heterogeneity attributable to differences in sample size, ethnicity, environment and study design, and identified low-income environments and ambient humidity as contributory risk factors [19]. A large cross-sectional study of elite summer-sport athletes found an overall EIB prevalence of around twenty-eight per cent, highest in swimmers and lowest in wrestlers, higher in females than males and higher in outdoor than indoor disciplines, and associated with elevated inflammatory biomarkers—evidence both of the scale of the problem and of its environmental and biological determinants [20]. The epidemiological dominance of EIB over EILO has an important clinical corollary: it creates a diagnostic bias whereby clinicians reflexively attribute any exertional wheeze to the lower airway, and this statistical anchoring is one of the principal reasons that the less common but frequently coexisting EILO is overlooked [3,19].

In children the epidemiology of EIB is shaped strongly by asthma and atopy. Among children and adolescents with asthma the prevalence of EIB is high, commonly cited between forty and ninety per cent and greatest in severe or poorly controlled disease, whereas in the general paediatric population estimates range from about five to twenty per cent, with considerable geographical and methodological variation [5,9,10]. A systematic review of adolescent athletes with asthma reported EIB prevalences spanning roughly two to sixty per cent, and consensus documents emphasise that EIB is frequently under-diagnosed in young athletes because symptoms are concealed, because self-reported dyspnoea is unreliable, and because objective challenge testing is not routinely performed [10,19]. Allergic rhinitis, atopic dermatitis and sensitisation to indoor allergens are consistent risk factors, so that the assessment of the breathless child should include a search for atopy, while recognising that a non-atopic, sports-asthma phenotype also exists and that neither atopy nor its absence resolves the differential with

EILO [5,9].

The biological correlates of EIB in athletes reinforce its inflammatory and environmental nature. In the large cohort of elite summer-sport athletes, those with EIB showed pronounced post-exercise falls in FEV1 together with elevated eosinophils, neutrophils and a range of inflammatory mediators compared with their EIB-negative counterparts, indicating that the condition is accompanied by measurable airway and systemic inflammation rather than being a purely functional phenomenon [20]. Winter-sport athletes are a particularly high-risk group: systematic evaluation of cross-country skiers and ice-hockey players has documented immune activation, systemic inflammation and airway dysfunction attributable to repeated exposure to cold, dry and, in ice sports, pollutant-laden air, with a burden of airway disease that rises with the number of years in competition [14]. Swimmers, exposed to chlorine disinfection by-products, likewise show a high frequency of airway epithelial injury and EIB, although the net effect of the pool environment remains debated [16]. These sport-specific patterns are epidemiologically informative and clinically useful for risk stratification, but—as with the symptom-based approach—they cannot establish the diagnosis in the individual athlete, which still requires objective bronchoprovocation [14,20].

7. Clinical Presentation and the Overlap of Symptoms

The clinical presentations of EILO and EIB overlap sufficiently to defeat diagnosis by history alone, and it is precisely this overlap that generates the breathless-athlete conundrum [2]. Both conditions produce exertional dyspnoea, chest tightness and audible respiratory noise, and both may be accompanied by cough and by a subjective sense that breathing has simply become harder [1,2]. Athletes with EILO characteristically report symptoms during the inspiratory phase of breathing, occurring when ventilatory demand is at its greatest—that is, at or near peak exercise intensity—and frequently describe throat or upper-chest tightness, a choking sensation, noisy breathing or frank stridor, hoarseness and, in severe episodes, panic [1]. A defining temporal feature is that the symptoms of EILO resolve rapidly, usually within a few minutes of stopping exercise, unless hyperventilation continues [1].

Symptoms attributable to EIB follow a different temporal and phasic pattern. Patients with EIB can usually continue exercising reasonably well while symptomatic, experience mainly expiratory difficulty and expiratory wheeze, and typically reach their worst breathlessness some minutes after exercise has ceased—commonly ten to fifteen minutes afterward—reflecting the delayed kinetics of mediator-driven bronchoconstriction [1,9]. The quality of the respiratory sound is a valuable, if imperfect, discriminator: EILO tends to produce a high-pitched inspiratory stridor arising from the larynx, whereas EIB produces an expiratory, lower-pitched wheeze arising from the bronchi [1]. In practice, however, athletes are often unable to localise their sensations or to identify the phase of breathing affected, and they may use terms such as throat tightness and chest tightness interchangeably, so that these distinctions, although real and clinically useful, cannot be relied upon in isolation [1,3].

The clinical challenge is compounded by the diversity of symptom description between individuals, which is coloured by prior experience, ambition and lifestyle, and by the fact that some athletes with objectively confirmed EILO deny ever having stridor [1,2]. In a large series of athletes with EILO, a substantial minority reported no stridor at all, a reminder that the absence of the classic laryngeal sign does not exclude laryngeal disease [2]. Finally, the differential diagnosis of exertional dyspnoea extends beyond EILO and EIB to encompass dysfunctional breathing and breathing-pattern disorders, sinonasal disease, cardiac and vascular anomalies, physical deconditioning, and rarer structural or neuromuscular conditions, several of which can coexist with laryngeal or bronchial disease and further muddy the clinical picture [1,6]. For all these reasons, caution must be exercised in making diagnostic decisions on the

basis of symptoms alone, and a systematic, objective approach is required [1,2].

8. Why Symptoms and Questionnaires Cannot Separate the Two Conditions

The inadequacy of symptom-based diagnosis is not merely a clinical impression but a finding supported by direct evidence. Guidelines for asthma and EIB have long advocated objective diagnostic testing precisely because a diagnosis of EIB based solely on patient-reported symptoms leads to frequent error, yet in routine practice such testing is often omitted, with the result that other treatable conditions are overlooked and asthma is both over- and under-diagnosed [1,2]. When athletes have been questioned qualitatively about their perception of exercise-induced symptoms, the correspondence between what they report and what objective testing demonstrates has been poor, and self-reported exertional dyspnoea has repeatedly proved to be a weak predictor of both EIB and EILO [17].

The same limitation applies to structured questionnaires, even those designed for laryngeal disease. In a study of a symptomatic, largely asthmatic population undergoing laryngoscopy, no combination of symptoms and no validated questionnaire—including instruments developed for vocal cord dysfunction, for dysfunctional breathing and for the differentiation of asthma from VCD—could reliably confirm or exclude laryngoscopy-proven ILO; dysphonia emerged as a modest clue, but candidate symptoms traditionally considered typical, such as throat tightness, were unhelpful individually or in combination [21]. The authors concluded that the diagnosis of VCD/ILO cannot be made confidently from clinical symptoms and questionnaires and that laryngoscopy remains essential, and they doubted that further refinement of symptom-based strategies would ever yield a validated rule-in or rule-out tool [21]. A parallel scoping review of outcome measurement in ILO similarly found that the instruments in use are heterogeneous and dominated by patient-reported measures that capture the burden of disease but cannot, by themselves, establish the diagnosis or distinguish laryngeal from bronchial pathology [40].

The final and perhaps most decisive argument against symptom-based diagnosis is the demonstration that laryngeal obstruction of a grade compatible with EILO occurs in a third of asymptomatic healthy athletes and that, conversely, symptoms of exertional dyspnoea are common in athletes with entirely normal laryngeal and bronchial findings [13,17]. Neither the presence of symptoms nor their character can therefore be equated with the presence or the nature of airway obstruction. The logical consequence is unavoidable: the two conditions can be reliably separated only by objective tests that visualise the larynx during exercise on the one hand and demonstrate bronchoconstriction on the other, and any diagnostic pathway for the breathless athlete that stops at the history and the questionnaire is fundamentally insufficient [2,13,21].

The clinical implication of this evidence is not that history-taking is worthless but that it must be positioned correctly within the diagnostic pathway. A carefully taken history—attending to the phase of breathing affected, the timing of symptoms relative to exercise, the character and localisation of any noise, the response to previous asthma treatment, and the presence of triggers such as cold air, chlorine or strong scents—generates the clinical suspicion that determines which objective test to perform, and video recordings of a characteristic field episode captured on a smartphone can meaningfully complement the history [1,2]. What the history cannot do is confirm the diagnosis, and the skilled clinician therefore uses it to direct objective testing rather than to replace it; the high rate of confirmed EILO among athletes referred by well-informed practitioners who recognised the clinical pattern demonstrates that expert clinical suspicion, when it triggers the correct test, is a powerful diagnostic asset [2].

9. Diagnostic Approach to EILO

The diagnostic gold standard for EILO is the continuous laryngoscopy exercise (CLE) test, in which a flexible video-laryngoscope is passed transnasally and secured so that the larynx can be visualised continuously throughout a maximal exercise test, most often on a treadmill and frequently combined with cardiopulmonary exercise testing [1]. Because EILO is a dynamic phenomenon that appears at the highest levels of ventilation and resolves within minutes of stopping, laryngoscopy performed only before and after exercise is inferior to continuous visualisation, which permits the severity of laryngeal closure to be assessed at the precise moment the athlete is most symptomatic [1]. The test also provides a direct biofeedback opportunity, allowing the athlete to see the larynx closing during symptoms, and it is indispensable in selecting patients for surgery, since only obstruction demonstrated to be supraglottic should ever lead to a supraglottic operation [1]. Ideally the exercise protocol should be intense enough, and where possible sport-specific enough, to reproduce real-life symptoms and to drive ventilation to the levels at which EILO manifests [1].

In practice, the exercise protocol used for the CLE test must be intense enough to reproduce the conditions under which EILO appears, because the obstruction manifests only at the highest levels of ventilation; testing should therefore proceed to exhaustion or to intolerable symptoms rather than stopping at an arbitrary submaximal heart rate [1,13]. Treadmill running is generally preferred to cycle ergometry in young people because it more reliably drives ventilation to maximal levels, although bicycle, rowing, swimming and stair-climbing protocols have all been used, and where feasible the mode should be tailored to the sport and to the triggers identified from the history [1]. The importance of achieving true maximal effort is underlined by the concern that tests terminating at ninety per cent of predicted maximal heart rate may fail to provoke EILO in susceptible individuals and thereby underestimate its prevalence, a methodological point that helps to explain the wide variation in reported prevalence between studies [1,13]. A combined CLE and cardiopulmonary exercise test, which yields maximal-oxygen-uptake data comparable to those of a conventional cardiopulmonary test, allows the laryngeal response and the broader differential diagnosis of exertional dyspnoea to be assessed within a single examination [1].

Laryngeal obstruction observed during the CLE test is graded semi-quantitatively using the widely adopted scoring system, in which supraglottic and glottic structures are each scored from zero, denoting complete patency, to three, denoting near-complete closure, at moderate and at maximal exercise intensity; a score of two or more at either the glottic or the supraglottic level is generally regarded as a significant obstruction compatible with EILO [1,17]. This system, although the most used, is a partially subjective visual assessment, and studies of its validity have reached mixed conclusions, so that defining EILO from a threshold score remains a challenge and the boundary between a normal and an abnormal laryngeal response is not sharply drawn [1,13]. The recognition that a third of asymptomatic athletes reach a supraglottic score of two reinforces the point that the score must be interpreted in the light of symptoms rather than in isolation [13]. Emerging quantitative measures, most notably translaryngeal airway resistance, offer the prospect of an objective functional index to complement the subjective visual score; translaryngeal resistance can be measured reproducibly during maximal exercise and rises with increasing laryngeal obstruction in females, representing an incipient functional validation of the visual grading, although it is not yet available for routine use [12].

A recurring practical obstacle is that the CLE test requires a comprehensive exercise laboratory, specialised equipment and skilled personnel, and is not available in most healthcare settings; this scarcity contributes materially to diagnostic delay and to the economic burden borne by patients with EILO [1,41]. Several groups have responded by adapting the method to make it more widely deliverable. A low-cost, self-manufactured head-mounted device combined with

a disposable flexible laryngoscope has been shown to allow CLE testing in children, confirming EILO and simultaneously helping to exclude EIB when serial spirometry is added, albeit with mild side effects such as tearing and vagal symptoms and a need to adjust the headgear during the test [22]. Investigators have also demonstrated the feasibility of performing the CLE test while the athlete breathes cold, dry air, reproducing the environmental conditions of winter sport and revealing increased glottic obstruction under cold-air challenge in preliminary work, an approach that may help to explain the anecdotal worsening of EILO in cold environments [23]. At the far end of technical sophistication, continuous bronchoscopy during exercise has been performed to interrogate the large airways below the larynx, and a paediatric case report combining continuous laryngoscopy, cardiopulmonary exercise testing and continuous bronchoscopy diagnosed supraglottic EILO while simultaneously excluding tracheomalacia in a child with a vascular ring—an illustration of how far dynamic exercise endoscopy can be extended to resolve complex exertional dyspnoea [24].

Alternative attempts to diagnose EILO without exercise have generally proved unsatisfactory. Resting spirometry and resting flow-volume-loop analysis lack face validity in a condition that by definition does not obstruct the airway at rest, and empirical studies confirm a poor correlation between resting spirometry and CLE findings [1]. Bronchial provocation tests other than exercise, and eucapnic voluntary hyperpnoea performed with or without concurrent laryngoscopy, have been explored as diagnostic short-cuts but have yielded divergent conclusions, and because the physiological breathing responses that evolve naturally during exercise do not appear during voluntary hyperpnoea, symptoms experienced while exercising are best interrogated by a test that also involves exercise [1]. The position of eucapnic voluntary hyperpnoea in the diagnostic hierarchy for EILO therefore remains unsettled, and where the history and clinical impression are not clear, a CLE test is required to establish the diagnosis definitively and to characterise the obstruction as glottic or supraglottic, a distinction that directly governs treatment [1].

Beyond the standard visual score, several quantitative scoring approaches have been developed in an effort to reduce subjectivity, including estimation of the anterior glottic angle or anterior-posterior diameter, a computerised pixel-counting method, and machine-learning analysis of laryngoscopic video using convolutional neural networks [1]. None has yet displaced the visual CLE score in routine practice, and studies of the reproducibility and validity of grade scoring have reached variable conclusions, reflecting the inherent difficulty of quantifying a continuous, dynamic process from endoscopic images [1]. The reproducibility of the CLE test as a whole is nonetheless reasonable in experienced hands: in the randomised crossover trial of ipratropium, in which each participant performed several CLE tests, the scores remained similar across tests, suggesting that the laryngeal response pattern of an individual is relatively stable and that experienced assessors can grade EILO with acceptable consistency [31]. These developments matter for the differential diagnosis because a more objective and reproducible measure of laryngeal obstruction would sharpen the boundary between the athletic and the pathological larynx and would strengthen the interpretation of treatment effects [12,31].

10. Diagnostic Approach to Exercise-Induced Bronchoconstriction

The diagnosis of EIB rests on the objective demonstration of exercise-provoked airflow limitation, because clinical assessment—history, examination and even video review—cannot accurately predict its presence or severity [9,10]. Baseline spirometry is usually normal in athletes with EIB and cannot by itself confirm the diagnosis; the essential step is to document a fall in forced expiratory volume in one second (FEV₁) after a standardised challenge [9,10]. The standardised treadmill exercise-challenge test is the most widely adopted method: the athlete exercises to reach a high proportion of maximal heart rate or target ventilation within

two to three minutes and sustains it for at least six minutes, after which FEV1 is measured at intervals, typically at five, ten, fifteen and thirty minutes [9,10]. A fall in FEV1 of at least ten per cent from baseline is diagnostic of EIB, with the severity graded as mild, moderate or severe according to the magnitude of the decline; a free-running test may be used in younger or less cooperative children, generally with a higher diagnostic threshold [9,10]. The test environment matters, and challenges are ideally conducted in cool, dry, controlled conditions to maximise sensitivity [9,10].

Because a single exercise challenge may be falsely negative and because athletes may respond to only one type of stimulus, indirect bronchoprovocation tests are frequently employed, particularly in the athletic setting [5,9]. Eucapnic voluntary hyperpnoea, in which the participant voluntarily ventilates dry air enriched with carbon dioxide at a high target rate for several minutes, is regarded as the reference indirect challenge for EIB in athletes and is incorporated into anti-doping assessment; a fall in FEV1 of at least ten per cent is diagnostic [5,9]. Osmotic challenges with inhaled mannitol or hypertonic saline provide alternative indirect methods, causing bronchoconstriction through the release of endogenous mediators, with a positive result usually defined by a fifteen per cent fall in FEV1 [5,9]. Direct challenges such as methacholine test bronchial hyper-responsiveness but are generally less effective than indirect challenges at detecting the exercise-related bronchoconstriction of the elite athlete, because indirect stimuli more closely reproduce the osmotic events of exercise [5,9]. Fractional exhaled nitric oxide can support a diagnosis of eosinophilic airway inflammation and may help identify athletes likely to have EIB, but it lacks the sensitivity to replace bronchoprovocation testing [5,9]. Importantly, athletes with EIB without asthma often exhibit normal exhaled nitric oxide, limited response to inhaled corticosteroids and few allergic features, reflecting the non-eosinophilic sports-asthma phenotype [5].

A crucial methodological point that has direct diagnostic relevance for the differential with EILO is the appearance of the flow-volume loop and of the laryngoscopic picture. Whereas EIB produces a fall in expiratory flow, laryngeal obstruction characteristically flattens the inspiratory limb of the flow-volume loop; yet flow-volume loops obtained during standard bronchoprovocation cannot reliably diagnose EILO, and patients with EILO frequently have entirely normal loops and normal spirometry [1,4]. It follows that a normal bronchoprovocation test and normal spirometry in a symptomatic athlete do not exclude an upper-airway cause and should, on the contrary, raise the suspicion of EILO and prompt referral for a CLE test [2,4]. The two diagnostic pathways are therefore complementary rather than competing: bronchoprovocation interrogates the lower airway, continuous laryngoscopy interrogates the upper airway, and in the athlete with unexplained exertional dyspnoea both may be required, ideally not on the same day to avoid the confounding effects of repeated maximal exercise [1].

Where the exercise challenge is impractical or inconclusive, standardised indirect challenges extend the diagnostic repertoire. The inhaled mannitol test, a dry-powder osmotic challenge, is safe and standardised and detects EIB through endogenous mediator release, with a positive result usually defined by a fifteen per cent fall in FEV1; it requires only a spirometer, has a single operating protocol, and its main drawbacks are cough and transient discomfort [9]. Hypertonic saline provides an alternative osmotic stimulus, and eucapnic voluntary hyperpnoea remains the reference indirect challenge in athletes and is embedded in anti-doping assessment [5,9]. A stepwise diagnostic algorithm for the young athlete typically begins with a symptom history and atopy assessment, proceeds to baseline spirometry with a bronchodilator to identify overt asthma, and then to an objective challenge—exercise or an indirect surrogate—to confirm or exclude EIB, with fractional exhaled nitric oxide offering supportive but non-definitive evidence of eosinophilic inflammation [5,9]. Because self-reported dyspnoea is a weak predictor and because athletes frequently conceal symptoms, screening based on symptoms

alone will substantially under-detect EIB, and objective testing during periods of intensive training is recommended [5,19].

Cardiopulmonary exercise testing has a valuable role when the diagnosis is uncertain, when the respiratory function tests are inconclusive, or when a non-respiratory cause of exertional dyspnoea is suspected, because it distinguishes ventilatory from cardiac limitation and identifies physical deconditioning [9,10]. A deconditioned athlete characteristically shows a reduced anaerobic threshold or a reduced peak oxygen uptake at the onset of symptoms together with an excessive heart-rate response for a given workload, whereas an athlete with a physiological limitation reaches a normal or elevated peak oxygen uptake [9,10]. Because deconditioning, dysfunctional breathing and cardiac disease can all masquerade as exercise-induced asthma, and because a combined continuous laryngoscopy and cardiopulmonary exercise test permits the larynx, the lungs and the cardiovascular response to be evaluated simultaneously, the assessment of the breathless athlete is increasingly conceived as an integrated, whole-system exercise evaluation rather than a single-organ test [1,9].

11. Coexistence of EILO and Asthma

A central complication of the differential diagnosis is that EILO and asthma are not mutually exclusive but frequently coexist, so that identifying one does not exclude the other and may leave the second condition untreated [1,2]. Laryngeal abnormalities have been reported to accompany difficult-to-treat asthma in a significant proportion of patients, and the relationship between EILO and the asthma paradigm continues to be debated [25]. Estimates of the prevalence of EILO among asthma patients vary from roughly thirteen to forty per cent, with the higher figures derived largely from specialist centres treating more severe disease, and a systematic meta-analysis has reported a pooled prevalence of comorbid laryngeal dysfunction in asthma of around thirty-eight per cent [25]. Because eight million people in the United Kingdom alone carry a diagnosis of asthma, even a conservative co-prevalence implies that a large number of individuals live with unrecognised laryngeal obstruction superimposed on their bronchial disease [25].

The association is strongest in difficult-to-treat and severe asthma, in which a high prevalence of laryngeal obstruction during exercise has been reported and in which unrecognised EILO may account for a proportion of the persistent symptoms that drive treatment escalation [25]. This has prompted the proposition that the larynx should be routinely considered in the assessment of asthma that responds poorly to guideline-based therapy, and prospective studies are now examining the prevalence of EILO in asthma, the effect of treating comorbid EILO on asthma control and quality of life, and the burden of other shared comorbidities such as obstructive sleep apnoea, gastro-oesophageal reflux, sinonasal disease and anxiety [25]. Until these data mature, the pragmatic clinical inference is that difficult asthma warrants a search for an upper-airway contribution before treatment is intensified, since escalating anti-inflammatory therapy cannot relieve a mechanical laryngeal obstruction and may expose the athlete to avoidable side effects [25].

The clinical importance of this coexistence lies in its capacity to generate treatment failure and diagnostic misattribution. In a study of national- and international-level athletes with confirmed EILO, coexisting objective evidence of asthma was common, and a majority had used asthma medication at some time, yet only a very small number reported that asthma medication had improved their exercise-related breathing problems, and of the few who did benefit almost all still required specific treatment for EILO [2]. This pattern—persistent exertional symptoms despite adequate asthma therapy—is one of the strongest clinical signals that an upper-airway component is present, and it should prompt referral for laryngeal assessment rather than escalation of bronchodilator or corticosteroid treatment [2]. In cohorts of patients referred for

exercise-induced dyspnoea, those diagnosed with EILO were significantly younger and less likely to carry a self-reported asthma diagnosis than those without EILO, but a meaningful minority used asthma medication despite no asthma symptoms, illustrating how readily laryngeal symptoms are misattributed to the lower airway [18]. Whether the treatment of comorbid EILO improves asthma control has not been systematically established, and this remains an open question that ongoing prospective studies are designed to address [25].

The reverse scenario—laryngeal symptoms driven by lower-airway or extra-laryngeal disease—also occurs and must be considered. A case report of a woman labelled as difficult-to-control asthma illustrates the point vividly: her cough, hoarseness and exertional breathlessness were unresponsive to high-dose inhaled therapy, her objective asthma tests were unremarkable, and her symptoms resolved only when vocal cord dysfunction, associated in her case with acid reflux, was recognised and treated with reflux management and speech therapy for vocal cord relaxation, after which she was able to stop asthma medication entirely and a subsequent methacholine challenge was negative, excluding asthma [26]. Such cases demonstrate both that laryngeal dysfunction can masquerade as severe asthma and that a wrong initial diagnosis can commit a patient to years of unnecessary and escalating treatment, reinforcing the imperative to confirm the diagnosis objectively and to exclude asthma mimics when asthma appears refractory [26].

12. The Broader Differential Diagnosis of the Breathless Athlete

Although EILO and exercise-induced asthma dominate the differential diagnosis of exertional dyspnoea in athletes, they are not its only causes, and a rigid focus on the upper and lower airways risks overlooking conditions that require entirely different management [1,9]. The symptoms of dyspnoea, cough, wheeze and chest tightness are not specific to any single diagnosis, and a systematic approach that keeps the broader differential in view is essential, particularly when the airway assessment is negative or the response to airway treatment is disappointing [9,10].

Structural and vascular anomalies of the large airways can produce exertional symptoms that mimic both EILO and EIB. Tracheomalacia—excessive collapsibility of the trachea, sometimes secondary to a vascular ring—can cause exercise-provoked cough, wheeze and dyspnoea, and is best assessed dynamically during exercise rather than with forced expiratory manoeuvres, which generate non-physiological pressures and may produce false-positive collapse; the feasibility of continuous bronchoscopy during exercise has recently been demonstrated in adults and applied in a paediatric patient in whom it distinguished supraglottic EILO from tracheomalacia caused by a double aortic arch [24]. A flattened inspiratory limb of the flow-volume loop suggests extrathoracic obstruction, whereas a flattened early-expiratory limb suggests central intrathoracic obstruction such as a vascular ring, prompting cardiac imaging [9].

Cardiac causes—arrhythmias, cardiomyopathies, shunts, valvular disease and pulmonary hypertension—must be considered, especially where dyspnoea is accompanied by chest pain or palpitations or where there is a family history of sudden cardiac death, and cardiopulmonary exercise testing with electrocardiography is the appropriate screening framework [9]. Metabolic and neuromuscular disorders, although rare, can present with exertional fatigue and breathlessness, and physical deconditioning is a common and frequently overlooked explanation for exercise intolerance even in trained individuals [9,10]. Finally, dysfunctional breathing and breathing-pattern disorders—including hyperventilation, thoracic-dominant breathing and periodic deep sighing—are common, frequently coexist with airway disease, and may sustain symptoms after the primary airway problem has been treated; their recognition matters because they respond to breathing retraining and, where appropriate, psychological

support rather than to airway medication [6,30,42]. The frequent coexistence of these conditions with EILO and EIB is precisely why the breathless athlete is best served by a structured, multidisciplinary evaluation rather than by empirical airway treatment [1,2,6].

13. Differentiating EILO from EIB: A Structured Comparison

Bringing together the features discussed above, the two conditions can be distinguished by a coherent constellation of anatomical, physiological, temporal, acoustic and therapeutic characteristics, provided that these are interpreted collectively and confirmed objectively rather than applied as isolated rules [1,2,3]. The site of obstruction is the larynx in EILO and the bronchi in EIB; the mechanism is a mechanical, aerodynamically driven, suction collapse in EILO and an inflammatory, mediator-driven bronchoconstriction in EIB; the phase of breathing predominantly affected is inspiration in EILO and expiration in EIB; the timing of maximal symptoms is at peak effort with rapid resolution on stopping in EILO but characteristically after exercise with slower resolution in EIB; the typical sound is a high-pitched inspiratory stridor localised to the throat in EILO and an expiratory wheeze localised to the chest in EIB; and the response to a bronchodilator is minimal in EILO but usually significant in EIB [1,2,3]. The following table summarises these discriminating features, which are drawn from and consistent across the reviewed literature [1,2,3,9].

Feature	EILO (upper airway)	EIB / exercise-induced asthma (lower airway)
Site of obstruction	Larynx (mainly supraglottic; glottis often secondary)	Bronchi / lower airways
Underlying mechanism	Mechanical, aerodynamic (Bernoulli) suction collapse	Inflammatory bronchoconstriction and airway oedema
Breathing phase affected	Predominantly inspiratory	Predominantly expiratory
Onset of attack	Sudden, within seconds, at high intensity	Gradual, minutes into or after exercise
Timing of peak symptoms	During maximal effort	Typically 10-15 minutes after exercise
Resolution	Rapid (usually within a few minutes of stopping)	Slower (often 20-60 minutes without inhaler)
Typical sound	High-pitched inspiratory stridor	Expiratory wheeze
Location of sensation	Throat / upper chest	Mid-to-lower chest
Response to salbutamol	Minimal or none	Usually significant
Baseline / resting tests	Normal spirometry; normal larynx at rest	Spirometry may be normal at rest
Diagnostic gold standard	Continuous laryngoscopy during exercise (CLE)	Exercise or indirect bronchoprovocation (fall in FEV1)
First-line treatment	Non-pharmacological (breathing retraining, speech therapy, IMT); surgery in selected supraglottic cases	Inhaled corticosteroids and beta-2 agonists

Table 1. Principal features distinguishing exercise-induced laryngeal obstruction from exercise-induced bronchoconstriction. Adapted and synthesised from the reviewed literature [1,2,3,9]. FEV1,

forced expiratory volume in one second; IMT, inspiratory muscle training.

Two caveats must accompany any such table. First, no single feature is decisive, because athletes localise and describe symptoms inconsistently, stridor may be absent in genuine EILO, and cold, dry air can aggravate both conditions [1,2]. The table is therefore a guide to clinical suspicion, not a substitute for objective testing. Second, the conditions coexist often enough that the demonstration of one should never be taken to exclude the other; the athlete whose objective asthma is controlled but who remains symptomatic at peak exercise is the paradigmatic candidate for laryngeal assessment [2,25]. Used in this spirit—as a framework for structuring suspicion that is then confirmed by continuous laryngoscopy for the upper airway and bronchoprovocation for the lower—the comparison provides a rational basis for separating two conditions whose surface presentations are deceptively similar [2,3].

In practical terms, a small number of clinical red flags should prompt specific consideration of EILO in an athlete otherwise assumed to have asthma: inspiratory rather than expiratory difficulty, symptoms that are maximal at peak effort and resolve within minutes of stopping, a sensation localised to the throat, audible inspiratory stridor, a poor or absent response to inhaled bronchodilators, and—perhaps most importantly—the persistence of exertional symptoms despite adequate, objectively justified asthma treatment [1,2]. None of these features is individually diagnostic, but their presence should redirect the diagnostic pathway toward laryngeal assessment rather than toward further intensification of asthma therapy, and their systematic elicitation is a low-cost means of reducing the diagnostic delay that so characterises EILO [2]. Conversely, expiratory wheeze that peaks after exercise, chest rather than throat localisation, a clear bronchodilator response and objective evidence of bronchial hyper-responsiveness point toward EIB and justify a lower-airway focus, while the coexistence of both symptom patterns should raise the possibility that both conditions are present [1,2].

14. Management of EILO

The management of EILO is predominantly non-pharmacological, a fact that follows directly from its mechanical rather than inflammatory nature and that sets it apart categorically from the management of asthma [1]. The overarching aim of conservative treatment is to teach the athlete to develop better control of the larynx during exercise so that an adequately open airway is maintained at high ventilation, and the principal modalities in widespread use are structured breathing advice with biofeedback, speech therapy or laryngeal control therapy, and inspiratory muscle training, with supraglottoplasty reserved for selected severe supraglottic cases [1]. A critical limitation of the evidence base, repeatedly emphasised by expert reviewers, is that no adequately powered randomised controlled trials of these interventions have been completed, so that most treatment recommendations rest on observational studies, single-centre case series and expert opinion, and comparative efficacy between modalities is largely unknown [1].

An important principle that unifies the conservative and surgical management of EILO is that the two major subtypes—predominantly glottic and predominantly supraglottic obstruction—probably require different therapeutic emphases and must be distinguished by the CLE test before treatment is chosen [1]. Speech therapy and breathing retraining appear to act most effectively at the glottic level, inspiratory muscle training has been associated principally with improvement in glottic scores, and surgery is appropriate only for supraglottic obstruction; applying the wrong modality to the wrong subtype is a plausible cause of treatment failure [1,27,28]. This subtype-specific logic is one of the strongest practical arguments for insisting that the CLE test not merely confirm the diagnosis but characterise the level of obstruction, and it distinguishes the individualised management of EILO from the more uniform, guideline-driven pharmacotherapy of asthma [1].

Breathing advice with biofeedback is the standard first-line treatment at experienced centres and exploits the CLE test itself: after the diagnosis is made, the laryngoscope is left in place while the athlete is taught breathing techniques and watches the effect on the larynx in real time, thereby learning consciously to abduct the laryngeal structures and to recognise and abort incipient closure [1]. The techniques emphasise diaphragmatic breathing, relaxation of laryngeal and shoulder-girdle tension, optimal posture, and specific manoeuvres to slow and control inspiratory flow, and the biofeedback element—seeing the larynx open in response to correct technique—is thought to accelerate learning [1]. Because any new breathing pattern must be repeated until it becomes automatic, the approach requires time, persistence and motivation, and its success is anecdotally greater in athletes who seek help early than in those who have had symptoms for years [1]. Prospective controlled evaluation of this widely used strategy is ongoing but not yet complete [25].

Speech therapy, sometimes termed laryngeal control therapy, is regarded in many countries as a first-line treatment for ILO and EILO, and encompasses education about the condition, establishment of lower thoracic and diaphragmatic breathing, relaxation training, and rehearsal of rescue techniques to manage symptoms [27]. A standardised five-session speech-therapy protocol evaluated with the CLE test in patients with EILO produced a statistically significant reduction in the visually scored laryngeal obstruction, with the greatest improvement at the glottic level, together with self-reported symptom improvement in eighty-six per cent of patients and a moderate correlation between the objective and subjective outcomes—the first study to demonstrate an effect of standardised speech therapy on visually determined laryngeal obstruction rather than on symptoms alone [27]. This work is notable both for its use of an objective endpoint and for its finding that standardisation, although it may sacrifice some individual tailoring, enhances reproducibility and accessibility; the authors appropriately caution that a single-centre, retrospective, uncontrolled design cannot exclude a contribution from reassurance or from the biofeedback provided at diagnosis, and that controlled trials are required [27].

The content of standardised speech-therapy protocols illustrates how conservative treatment is operationalised. A representative programme delivered over five sessions progresses from techniques that generate positive airway pressure and reduce laryngeal tension, through exercises that desensitise the patient to the sensation of laryngeal adduction and teach the difference between adduction and abduction, to the rapid, controlled switching from adduction to abduction and the graded application of these techniques during physical activity of increasing intensity [27]. The reprogramming of the muscles involved, the desensitisation to triggers, and the establishment of diaphragmatic breathing to support laryngeal patency are the recurring principles, and mastery of each set of exercises is required before the next is introduced, with substantial home practice between sessions [27]. The demonstration that such a programme reduces objectively scored laryngeal obstruction, and not merely symptoms, marks an advance over earlier studies that relied solely on self-report, although the absence of a control arm means that a contribution from reassurance and from the biofeedback provided at diagnosis cannot be excluded [27].

Inspiratory muscle training has been proposed as a treatment that primarily benefits the glottic component of EILO, on the rationale that strengthening and better coordinating the inspiratory musculature—exploiting the phasic relationship between the diaphragm and the posterior cricoarytenoid abductor—may help maintain laryngeal patency [28]. A study of patients treated with information and breathing advice plus six weeks of inspiratory muscle training found that self-reported symptoms and laryngeal scores improved in most participants shortly after treatment, with the improvement associated with a reduction in glottic scores in particular; however, at four to six years of follow-up, symptoms had improved to similar levels in this

group and in a control group that had received breathing advice alone, so that no long-term added benefit from inspiratory muscle training could be demonstrated, full symptom resolution was rare, and the decline in symptoms over time appeared to be linked partly to a decline in physical activity rather than to cure of the condition [28]. A case report has suggested that inspiratory muscle training may even be counterproductive in severe supraglottic EILO, and it has accordingly been proposed that low inspiratory resistance be used to target laryngeal coordination rather than muscle strength [1,28]. These findings temper enthusiasm and underline that treatment effects may not distinguish specific interventions from the natural history and from non-specific reassurance [28].

Broader multidimensional and physiotherapy-based approaches have also been described, reflecting a conceptual shift from a purely local view of the larynx to a whole-body view of breathing. A multidimensional intervention combining Norwegian psychomotor physiotherapy, elements of cognitive behavioural therapy and an individually tailored rehabilitation plan reduced inspiratory distress and dysfunctional, thoracic-dominant breathing in adolescent athletes with EILO, promoting a more diaphragmatic breathing pattern and improved posture without measurable change in lung function—implying that the benefit derived from altered breathing mechanics and reduced tension rather than from any change in pulmonary physiology [30]. This line of work situates EILO within the broader field of dysfunctional breathing, where thoracic-dominant breathing and forward head posture are thought to raise the larynx and generate paradoxical laryngeal forces, and where restoring diaphragmatic breathing is a shared therapeutic goal [30]. It is consistent with the observation that dysfunctional breathing patterns are common even among elite athletes and are associated with less favourable respiratory mechanics, suggesting that breathing-pattern assessment and retraining may have a place in the management of exertional dyspnoea more generally [42].

The recognition that dysfunctional, thoracic-dominant breathing is common even in elite athletes and is associated with measurably poorer pulmonary mechanics provides a physiological rationale for breathing-pattern retraining as a component of EILO management [42]. Restoring a diaphragmatic pattern, releasing accessory-muscle tension and correcting a forward-head posture are thought to lower the larynx to a more neutral position and to reduce the paradoxical forces acting on it during high-flow inspiration, and these goals are shared across physiotherapy, speech therapy and the multidimensional interventions described above [6,30,42]. Because such retraining addresses a modifiable functional contributor rather than a fixed anatomical defect, it is applicable to the majority of patients with EILO and, unlike surgery, carries little risk, which supports its position as a first-line strategy alongside education and biofeedback [6,30].

A significant methodological advance in the field has been the recognition that non-pharmacological interventions for ILO have historically been poorly described, non-standardised and inadequately theorised, which has impeded both their evaluation and their dissemination. In response, a manualised, theory-based intervention—Upper Airway Control Therapy—has been developed for adults with ILO using established frameworks for complex-intervention design, comprising core components of education and empowerment and of reliever breath control, supporting components including biofeedback training and prevention methods, and a cross-cutting home-practice component, underpinned by an explicit set of behaviour-change techniques [29]. Although developed principally for non-exercise-induced ILO and not yet tested for clinical effectiveness, this work exemplifies the direction the field must take if conservative treatments are to be evaluated rigorously and delivered reproducibly [29]. The importance of standardisation is reinforced by evidence that laryngeal retraining therapy benefits patients with a clinical diagnosis of VCD irrespective of whether laryngeal adduction is visible at laryngoscopy, so that response to conservative treatment does not by

itself confirm or refute the diagnosis and outcome measurement must be interpreted with care [7].

Pharmacological treatment has, by contrast, little to offer in EILO, and the evidence against several proposed drug therapies is instructive. Because the mechanism is mechanical, anti-inflammatory inhalers directed at the lower airway are ineffective, and no pharmacological therapy has been shown convincingly to improve EILO [1]. Inhaled ipratropium bromide had been suggested on the basis of anecdotal reports and a putative vagal mechanism, but a randomised crossover trial in participants with CLE-confirmed EILO found no effect of inhaled ipratropium on the grade of laryngeal obstruction, on self-assessed dyspnoea or on exercise capacity, and the authors explicitly recommended against its use while encouraging objective testing for both asthma and EILO in athletes with exercise-related breathing problems [31]. This negative trial is valuable precisely because it applied objective outcome measures to a question that had previously rested on uncontrolled reports, and it provides a model for the rigorous evaluation that other EILO treatments still lack [31].

Surgical treatment, in the form of supraglottoplasty or laser-assisted aryepiglottoplasty, is reserved for carefully selected, highly motivated patients with moderate to severe obstruction arising specifically from the supraglottic structures, in whom conservative measures have failed and symptoms significantly impair quality of life or performance [1]. There is general agreement that surgery must never be undertaken without clear CLE evidence of supraglottic obstruction, and follow-up studies generally report favourable outcomes, with improvements in symptoms and, in some series, in peak oxygen uptake; complications are uncommon but have included excessive scarring and transient reduction in vocal fold motion [1]. However, no randomised trials of surgery have been performed, the procedure is not standardised, and a systematic review concluded that the level of evidence precludes definitive recommendations for or against surgery, so that methodologically robust prospective studies are needed [1]. Post-operative rehabilitation to relearn optimal breathing patterns is usually required, because the central tendency to maladaptive laryngeal movement may persist after the anatomical obstruction has been surgically relieved [1].

Finally, the natural history and prognosis of EILO temper therapeutic expectations. Studies from general populations indicate that untreated or under-treated EILO leads many individuals to reduce their physical activity, and long-term follow-up suggests that full symptom resolution is uncommon and that the apparent decline in symptoms over years may reflect a withdrawal from the exercise that provokes them rather than a cure [28]. This has a public-health dimension, since the condition affects young people at an age when physical activity often begins to decline, and its proper handling may help maintain a healthy level of activity [28]. It also argues for early diagnosis and individualised follow-up rather than a single course of treatment, and for a realistic account to patients that management aims at control rather than cure [28,41].

15. Management of Exercise-Induced Asthma

The management of EIB stands in deliberate contrast to that of EILO and follows the general principles of asthma care, combining pharmacological and non-pharmacological measures aimed at controlling inflammation, preventing bronchoconstriction and permitting full participation in sport [4]. Inhaled corticosteroids are the cornerstone of pharmacological therapy in athletes with persistent or frequent EIB, controlling the underlying airway inflammation and reducing exercise-induced bronchoconstriction, and are added when a short-acting beta-2 agonist is required more than occasionally; they may be used alone or, when control is insufficient, combined with a long-acting beta-2 agonist [4]. Short-acting beta-2 agonists provide rapid relief and can be taken shortly before exercise to prevent bronchoconstriction, but their use as sole therapy is discouraged because frequent or excessive use may induce tolerance

through beta-2 receptor desensitisation and reduce their protective effect; contemporary guidance favours as-needed inhaled corticosteroid-formoterol combinations over reliance on short-acting beta-2 agonists alone in mild disease [4]. Leukotriene receptor antagonists and mast-cell stabilising agents are further options, the former offering protection against bronchoconstriction provoked by pollutants, and inhaled corticosteroids remain central where an inflammatory phenotype is present [4]. A prospective controlled study in elite endurance athletes demonstrated that thirty days of individualised inhaled corticosteroid plus long-acting beta-2 agonist therapy abolished eucapnic-hyperpnoea-provoked bronchoconstriction in most and increased peak oxygen uptake, confirming both the reversibility of EIB and its performance cost when untreated, although this benefit occurred without change in the allergic inflammatory response [32]. The correct administration technique for inhaled medication, and appropriate selection and use of inhaler devices, are themselves important determinants of treatment success in obstructive airway disease and are frequently overlooked [34].

Non-pharmacological management complements drug treatment and is particularly relevant to the athlete. A structured warm-up before vigorous exercise reduces subsequent bronchoconstriction in a substantial proportion of individuals by inducing a refractory period, and high-intensity interval or variable-intensity warm-ups appear most effective [4,26]. Trigger avoidance, where feasible—minimising exposure to cold, dry air, to high pollutant or allergen loads, and to chlorine by-products—together with the use of heat-and-moisture-exchanging masks in cold conditions and attention to nasal breathing, can reduce the frequency and severity of symptoms, and comorbid conditions such as rhinitis and gastro-oesophageal reflux should be identified and treated [4,26]. It is essential to emphasise that EIB is not a reason to avoid sport; on the contrary, regular exercise improves cardiorespiratory fitness, reduces airway inflammation and enhances asthma control and quality of life, and the paradoxical avoidance of exercise by people with asthma perpetuates deconditioning and worsens their disease [36,37]. Aerobic training has been shown to improve asthma control and lung function and, in some studies, to reduce medication requirements, and Mediterranean- or plant-rich dietary patterns and maintenance of a healthy body weight are associated with better asthma outcomes, all of which support the integration of lifestyle measures into the care of the athlete with asthma [36,37,38]. In children, appropriate control of asthma reduces the frequency and intensity of EIB, and pharmacological and non-pharmacological measures are combined within a stepwise, guideline-based framework [5,33].

The mechanisms and practicalities of non-pharmacological prevention merit expansion. A pre-exercise warm-up, particularly of high or variable intensity, reduces subsequent bronchoconstriction in a substantial proportion of individuals by inducing a refractory period lasting one to two hours, attributed to tachyphylaxis of airway smooth muscle to bronchoconstrictor mediators [5,26]. Heat-and-moisture-exchanging masks and scarves warm and humidify inspired air in cold conditions, nasal rather than mouth breathing preserves the conditioning function of the upper airway, and avoidance of the most polluted or allergen-laden environments where feasible all reduce the provocation of bronchoconstriction [5]. Among adjunctive medications, leukotriene receptor antagonists offer protection against bronchoconstriction—including that provoked by pollutants—and mast-cell stabilising agents provide a further, if less potent, option, although both are generally inferior to inhaled corticosteroids where an inflammatory phenotype predominates [4,5]. These measures are complementary to, not substitutes for, the accurate diagnosis on which rational treatment depends, and none of them has any role in the management of EILO, whose provocation is mechanical rather than inflammatory [3,4].

The effectiveness of inhaled therapy for EIB depends not only on the choice of drug but on the correct use of the delivery device, an aspect of asthma care that is frequently neglected and that

has particular relevance to young athletes. Pressurised metered-dose inhalers require coordination of actuation with inhalation and are best used with a spacer, whereas dry-powder inhalers demand that the patient generate a sufficient peak inspiratory flow, which varies between devices and with disease severity, so that device selection should be matched to the individual's inspiratory capacity and technique verified and re-checked over time [34]. Poor inhaler technique is a common and remediable cause of apparent treatment failure, and its correction is a simple means of improving control before therapy is escalated [34]. Because the misattribution of persistent symptoms to inadequate asthma treatment is one of the mechanisms by which EILO comes to be over-treated, attention to inhaler technique also serves the differential diagnosis, ensuring that genuine non-response is not mistaken for under-dosing [2,34].

A consideration unique to the athletic population is anti-doping regulation, which shapes the pharmacological management of EIB in a way that has no parallel in EILO [4,35]. Inhaled corticosteroids are permitted, and inhaled beta-2 agonists are permitted within specified maximum doses provided the athlete can document a therapeutic indication, so that an objective diagnosis of asthma or EIB is not merely good clinical practice but a regulatory necessity for the competing athlete [4,35]. Because inhaled beta-2 agonists and corticosteroids are scrutinised for potential performance-enhancing effects, and because the misattribution of EILO to asthma can lead to unwarranted therapeutic use exemptions and to the prescription of medication an athlete does not need, the requirement for objective confirmation of EIB serves both the athlete's health and the integrity of competition [1,35]. This regulatory dimension provides yet another reason—beyond diagnostic accuracy and treatment efficacy—for insisting that exertional dyspnoea in the athlete be characterised objectively rather than treated empirically [1,35].

Because inhaled therapy for asthma is not free of consequence, the accurate confirmation of EIB before treatment carries a safety as well as a regulatory rationale. Long-term inhaled corticosteroid therapy in children has been studied for potential systemic effects, including markers of vascular endothelial dysfunction, and the balance of benefit and risk is optimised only when the medication is directed at genuine airway inflammation rather than prescribed empirically for misattributed laryngeal symptoms [15,33]. Inhaled beta-2 agonists and corticosteroids have also been scrutinised for effects on the heart and skeletal muscle relevant to both safety and anti-doping surveillance, so that their unnecessary use in an athlete who in fact has EILO exposes that athlete to avoidable pharmacological risk and administrative burden without any therapeutic gain [35]. These considerations converge on the conclusion reached throughout this review: that objective diagnosis is the precondition for treatment that is both effective and safe, and that the empirical treatment of exertional dyspnoea as asthma is difficult to justify once the ready availability of definitive laryngeal and bronchial testing is acknowledged [1,35].

16. Consequences of Misdiagnosis and the Problem of Awareness

The stakes of distinguishing EILO from EIB are high because misdiagnosis carries tangible costs for the athlete and for the healthcare system. When EILO is mistaken for asthma, the athlete may be committed to prolonged, escalating and ineffective courses of inhaled and sometimes oral medication, with attendant side effects, unnecessary expense, and—in the competitive athlete—the additional burden of doping discussions and unwarranted therapeutic use exemptions, while the true condition remains untreated and performance-limiting [1,2]. The reverse error, treating a bronchial disease as a laryngeal one, is less common but equally harmful, and the coexistence of the two conditions means that treating only one may leave the athlete symptomatic and the clinician puzzled [2,25]. In paediatric ILO, diagnostic delay is the norm rather than the exception: a large consecutive series reported a median interval of more

than two years from symptom onset to diagnosis, with the great majority of patients requiring multiple specialist referrals, a substantial proportion carrying a presumptive diagnosis of asthma despite pulmonary function tests inconsistent with asthma, and more than half being treated with bronchodilators—an eloquent illustration of how readily laryngeal disease is misattributed and how prolonged the resulting patient journey can be [41].

The costs of this diagnostic delay are both economic and clinical. Children and adolescents with induced laryngeal obstruction incur substantial expenditure through repeated consultations, investigations and medications accumulated before the correct diagnosis is reached, and the prolonged symptom burden may lead them to reduce or abandon their sport [41]. Median times from symptom onset to diagnosis measured in years, the requirement for multiple specialist referrals, and the frequent prior labelling as asthma despite normal or non-asthmatic pulmonary function tests together depict a system poorly configured to recognise a mechanical upper-airway disorder [41]. Shortening this pathway will require not only better access to the CLE test but a change in clinical reflex, so that the larynx is considered early and objectively rather than after asthma treatment has visibly failed—an argument that recurs throughout this review and that is supported by the demonstrated benefit of skilled, EILO-aware clinicians in achieving rapid and accurate diagnosis [2,41].

The burden of living with unrecognised or poorly managed laryngeal obstruction is considerable and extends well beyond the physical symptom. Qualitative research into the lived experience of ILO has documented a heavy psychosocial toll, including loss of autonomy, fear and anxiety about unpredictable attacks, social isolation, and the frustration of feeling disbelieved by healthcare professionals during years of misdirected care before the correct diagnosis was finally reached [39]. Patients describe the relief that a definitive diagnosis brings and the value of a personalised, multidisciplinary and patient-centred approach, findings that argue for earlier recognition and for outcome measures that capture the full impact of the condition on life rather than symptoms alone [39,40]. The economic burden of induced laryngeal obstruction in children and adolescents, arising from repeated office visits, procedures and medications before the diagnosis is made, adds a further dimension to the cost of delay [41].

Underlying much of this diagnostic failure is a simple lack of awareness, not only among the public but among the professionals who first encounter the breathless athlete. A survey of athletic coaches—who, because EILO symptoms typically arise at peak effort during training and competition, are often the first adults to witness them—found that fewer than a quarter had any awareness of EILO, that neither the sport coached nor the coach's age or experience predicted awareness, and that coaches were significantly more confident dealing with exercise-induced dyspnoea of unknown cause than with EILO specifically [38]. Encouragingly, coaches who were aware of EILO used more appropriate, condition-specific strategies, being less likely to recommend inhaler use or to advise athletes to push through symptoms, and coaches overwhelmingly expressed a desire for better training and better communication with athletes and families [38]. These findings identify education of coaches, athletic trainers and physical-education staff, alongside better interdisciplinary communication with speech pathologists and physicians, as a practical lever for improving the early recognition of EILO and shortening the diagnostic delay [38]. The contrast with the high recognition of asthma among the same coaches mirrors the wider clinical asymmetry whereby the lower airway is considered first and the larynx last [38].

The remedy for low awareness is not confined to the education of coaches but extends to the construction of interdisciplinary care pathways. The optimal management of laryngeal and lower-airway disorders in athletes draws on the complementary expertise of sport and respiratory physicians, otolaryngologists, speech and language therapists, physiotherapists and,

where indicated, psychologists, and the multidisciplinary team model has been associated with more efficient management and reduced healthcare utilisation in inducible laryngeal obstruction [6,40]. Embedding the larynx in the diagnostic algorithm of the breathless athlete, ensuring communication between the clinicians who diagnose and those who treat, and coordinating care with coaches and families are recurring recommendations across the literature, and they reflect a broader recognition that exertional dyspnoea is frequently multifactorial and rarely resolved by a single discipline acting alone [6,38,40]. The development of standardised, manualised interventions and of disease-specific outcome measures is a further prerequisite for such coordinated care, because it allows different practitioners to deliver and evaluate treatment consistently [29,40].

17. Special Populations and Modifying Factors

Several demographic and environmental factors modify the presentation and the differential diagnosis of exertional dyspnoea and deserve specific comment. Sex is among the most consistent: EILO shows a female predominance that emerges around and after puberty, plausibly related to the smaller inner diameter of the female larynx and to the higher translaryngeal resistance measured in women, and the highest prevalence of EILO in referred cohorts is found in females under eighteen [1,12,18]. It is important, however, not to allow this association to narrow clinical suspicion unduly, because EILO occurs in adolescent males and in adults of both sexes at clinically meaningful rates, and an exclusive focus on the young female athlete risks overlooking the diagnosis in other patients equally affected [18]. EIB likewise shows sex- and sport-related variation, with higher prevalence reported in females and in outdoor and endurance disciplines in some large cohorts [20].

Age is a second modifier. EILO characteristically presents in the peripubertal and adolescent years, a period during which the larynx is undergoing developmental change, and the paediatric presentation may be particularly vague and easily confused with the bronchial obstruction of viral infections and asthma [1,41]. Diagnostic tools developed for adults must be adapted for children, and the CLE test in particular requires modification of equipment and technique for younger patients, in whom tolerance is generally good but side effects such as tearing and vagal responses are more common [22]. The developmental context also matters for management, since establishing durable changes in breathing pattern and behaviour in an adolescent requires an approach that engages the athlete, the family and, ideally, the coach [30,38].

The paediatric context deserves particular emphasis because children are both especially likely to be affected by EILO and especially likely to be misdiagnosed. The clinical picture in children may be dominated by the wheeze and cough of intercurrent viral infection, which mimics the bronchial obstruction of asthma, and pubertal changes in the larynx add a further confounding influence [1,41]. Objective diagnosis in children is complicated by the limited availability of the CLE test and by the need to adapt equipment and technique for smaller and less cooperative patients, although purpose-built low-cost head-mounted devices have made the test feasible in paediatric practice and allow EILO to be confirmed while EIB is simultaneously excluded by serial spirometry [22]. Given the frequency of misdiagnosis, the long diagnostic delays and the psychosocial toll documented in young people with induced laryngeal obstruction, there is a strong case for raising awareness of EILO among paediatricians, school staff and coaches, and for a low threshold to investigate exertional dyspnoea that does not respond to asthma treatment with a test that visualises the larynx during exercise [22,38,41].

Environment is the third major modifier and, as already noted, is a source of diagnostic ambiguity as well as of disease. Cold, dry air provokes EIB through osmotic and thermal mechanisms and is a defining exposure of winter-sport athletes, in whom airway dysfunction accumulates over years of training; the same environmental conditions have been reported to

worsen EILO, so that a history of cold-air-provoked symptoms does not localise the problem to the upper or the lower airway [1,5,14,16]. Swimmers are exposed to chlorine by-products that injure the airway epithelium and are associated with a high prevalence of EIB, though the net effect of the pool environment remains contested [5,16]. Athletes on ice and outdoor endurance athletes face their own pollutant and allergen exposures [5,9]. Because these environmental factors influence both conditions and because the sports-asthma phenotype they generate is often non-eosinophilic and relatively corticosteroid-unresponsive, environmental history is valuable for understanding risk and for informing preventive measures but cannot substitute for objective testing in the individual athlete [5,14].

A final modifying consideration is the frequent coexistence of dysfunctional breathing and breathing-pattern disorders with both EILO and EIB. Thoracic-dominant, apical and rapid breathing patterns are common even in elite athletes and are associated with less favourable pulmonary mechanics, and such patterns may both mimic and aggravate laryngeal obstruction by increasing extrinsic tension on the laryngeal framework and by raising the larynx within the neck [30,42]. Dysfunctional breathing, hyperventilation and anxiety therefore form part of the differential diagnosis of the breathless athlete and may contribute to the persistence of symptoms after the primary airway problem has been addressed; their recognition is important because they respond to breathing retraining and psychological support rather than to airway medication, and because they illustrate once again that exertional dyspnoea is frequently multifactorial [6,30,42]. A systematic, whole-airway and whole-body assessment—rather than a reflexive attribution to asthma—is thus required to disentangle the contributions of the larynx, the bronchi, the breathing pattern and the mind in the individual athlete [2,6].

A cross-cutting theme relevant to all these populations is that exertional respiratory disease, whether laryngeal or bronchial, must not be allowed to drive the athlete out of sport. Untreated EILO leads many young people to reduce their physical activity, and the apparent improvement of symptoms over years may reflect this withdrawal rather than resolution of the condition, with adverse consequences for long-term health [28]. In asthma the same trap operates: fear of exercise-induced symptoms promotes a sedentary lifestyle that worsens fitness and disease control, whereas regular aerobic exercise improves cardiorespiratory fitness, reduces airway inflammation and enhances asthma control and quality of life, and appropriate diet and weight management further improve outcomes [36,37,38]. For both conditions, therefore, the goal of management is not avoidance of exertion but the accurate diagnosis and condition-specific treatment that allow the athlete to continue training and competing safely, a goal that is defeated whenever EILO is mistaken for asthma and treated with medication that cannot help it [2,28,36].

18. Discussion

The evidence reviewed here supports a clear and clinically actionable thesis: EILO and exercise-induced asthma are physiologically opposite conditions—one a mechanical collapse of the upper airway, the other an inflammatory narrowing of the lower airway—that present with deceptively similar symptoms, coexist more often than is generally appreciated, and can be reliably distinguished only by objective testing directed at the appropriate part of the airway [1,2,3]. The historical dominance of asthma in the sports-medicine imagination has produced a systematic diagnostic bias in which the lower airway is interrogated first and the larynx, if at all, only after asthma treatment has visibly failed, and this bias is the proximate cause of the prolonged diagnostic delays, the misdirected treatment and the psychosocial burden documented across the literature [2,38,39,41]. Correcting it does not require abandoning the assessment of asthma, which remains common and important, but rather expanding the diagnostic algorithm so that the larynx is considered from the outset in any athlete with

exertional dyspnoea, and objectively evaluated whenever symptoms persist despite adequate asthma therapy [1,2].

The reviewed evidence also invites a conceptual reframing of the relationship between the larynx and the lower airway. Rather than regarding EILO and asthma as competing diagnoses to be chosen between, it is more accurate—and more clinically useful—to regard the airway as a continuum from the nose to the bronchi, any part of which may contribute to exertional dyspnoea, and to recognise that laryngeal obstruction may be an integral part of the asthma paradigm as well as a discrete disorder in its own right [1,25]. This whole-airway view accommodates the frequent coexistence of the two conditions, the shared environmental triggers such as cold air, the overlap with dysfunctional breathing, and the observation that laryngeal abnormalities cluster in difficult asthma [1,25,30]. It also implies that the diagnostic evaluation of the breathless athlete should be organised around the airway as a whole, with objective interrogation of each level as clinically indicated, rather than around a premature commitment to a single anatomical culprit [1,2].

A recurring and instructive tension in the field concerns the relationship between objective findings and symptoms. On the one hand, the review has emphasised that symptoms and questionnaires cannot separate EILO from EIB and that objective tests are indispensable [17,21]. On the other hand, the demonstration that a third of asymptomatic healthy athletes have supraglottic laryngeal obstruction of a grade conventionally regarded as diagnostic shows that objective laryngeal findings, taken in isolation, are equally capable of misleading [13]. The resolution of this apparent paradox is that neither symptoms nor findings suffice alone: EILO is properly defined as the conjunction of visualised laryngeal obstruction with compatible symptoms, and its diagnosis therefore demands that the CLE test be interpreted in the clinical context rather than as a stand-alone verdict [13]. The same principle applies to treatment response, since laryngeal retraining benefits patients irrespective of visible adduction and since symptom improvement over time may reflect reduced physical activity rather than cure, so that outcome measurement in EILO must combine objective and subjective endpoints and must be interpreted against the natural history [7,28]. The field is beginning to develop the quantitative tools—translaryngeal resistance, computational modelling and standardised, theory-based interventions—that this more nuanced diagnostic and therapeutic framework requires [11,12,29].

The weaknesses of the current evidence base must be acknowledged frankly, because they bound the confidence with which any recommendation can be made. For EILO, the most serious limitation is the near-complete absence of randomised controlled trials of treatment; the widely used conservative interventions rest on observational data, single-centre series and expert opinion, comparative efficacy is unknown, and the one high-quality randomised trial in the field—of inhaled ipratropium—was negative [1,31]. The visual CLE scoring system, though the accepted standard, is partially subjective and of contested validity, and the threshold that defines pathological obstruction is not firmly established [1,13]. Much of the literature derives from a small number of specialised centres, predominantly in Northern Europe and North America, so that the global generalisability of prevalence estimates and management strategies is uncertain, and access to the CLE test is limited in most settings, sustaining under-diagnosis [1,22,41]. For EIB, although the diagnostic methods are better standardised, the heterogeneity of prevalence estimates, the distinct and relatively corticosteroid-unresponsive sports-asthma phenotype, and the interaction with anti-doping regulation all complicate management, and the long-term respiratory consequences of years of high-volume ventilation in adverse environments remain incompletely characterised [5,14,20]. Ongoing prospective studies designed to establish the prevalence and impact of EILO in asthma, and to evaluate the treatment of comorbid EILO with validated outcome measures, are therefore of considerable

importance [25].

Several priorities for research and practice emerge from this synthesis. Methodologically, the field needs adequately powered randomised controlled trials of the conservative treatments for EILO, using standardised, manualised interventions and validated, disease-specific outcome measures that capture the full physical, social and personal impact of the condition [27,29,40]. Clinically, it needs wider access to, and where necessary simplified or surrogate versions of, the CLE test, so that objective diagnosis is not confined to a handful of tertiary centres [22,23]. Educationally, it needs to raise awareness of EILO among coaches, athletic trainers, primary-care and sports-medicine clinicians, and to embed the larynx in the differential diagnosis of the breathless athlete, supported by better interdisciplinary communication among physicians, speech pathologists and physiotherapists [38]. And conceptually, it needs to embrace a whole-airway, whole-body and multidisciplinary model of exertional dyspnoea in which the larynx, the bronchi, the breathing pattern, the environment and the psychological context are assessed together rather than in isolation, because the reviewed evidence shows repeatedly that these factors coexist and interact [2,6,30]. Only such an integrated approach can convert the growing scientific understanding of EILO and EIB into the timely, accurate and condition-specific care that the breathless athlete requires.

For the practising clinician, several concrete recommendations follow from this synthesis. Any athlete presenting with exertional dyspnoea should undergo a structured history and, as a minimum, spirometry before and after a bronchodilator, with objective bronchoprovocation where asthma is suspected; the larynx should be considered from the outset, and continuous laryngoscopy during exercise arranged whenever symptoms are predominantly inspiratory, arise at peak effort, resolve rapidly on stopping, are accompanied by throat tightness or stridor, or persist despite adequate asthma therapy [1,2]. A diagnosis of asthma should never be made or escalated on the basis of symptoms alone, and a diagnosis of EILO should rest on the conjunction of laryngeal obstruction and compatible symptoms rather than on either in isolation [13,21]. When EILO is confirmed, management should be non-pharmacological and matched to the subtype of obstruction, with surgery reserved for selected severe supraglottic cases, and asthma medication prescribed only where EIB has been objectively demonstrated [1,27]. Above all, the coexistence of the two conditions should be actively considered, so that neither is left untreated in the athlete who has both [2,25].

19. Conclusions

Exercise-induced laryngeal obstruction and exercise-induced asthma are common, physiologically opposite and frequently coexisting causes of exertional dyspnoea in athletes that cannot be reliably distinguished by symptoms and are too often conflated in clinical practice. On the basis of the evidence reviewed, the following principal conclusions may be offered.

First, EILO is a mechanical, aerodynamically driven, predominantly supraglottic and inspiratory collapse of the upper airway that peaks at maximal effort and resolves within minutes, whereas EIB is an inflammatory, expiratory bronchoconstriction of the lower airway that typically peaks after exercise; this fundamental difference in mechanism dictates that their diagnosis and treatment diverge and that asthma medication has no rational role in EILO.

Second, neither symptoms nor questionnaires can separate the two conditions, and objective testing is indispensable: continuous laryngoscopy during exercise is the diagnostic gold standard for EILO, and standardised exercise or indirect bronchoprovocation is required to confirm EIB. Because laryngeal obstruction of a diagnostic grade occurs in asymptomatic athletes, the diagnosis of EILO rests on the conjunction of visualised obstruction and

compatible symptoms rather than on either alone.

Third, EILO and asthma coexist frequently, so that demonstrating one does not exclude the other; persistent exertional dyspnoea that fails to respond to adequate asthma therapy should prompt objective assessment of the larynx rather than escalation of asthma treatment.

Fourth, the management of EILO is predominantly non-pharmacological—breathing retraining with biofeedback, speech therapy and inspiratory muscle training, with supraglottoplasty reserved for selected severe supraglottic cases—and rests on weak, largely uncontrolled evidence, with inhaled ipratropium shown to be ineffective; the management of EIB follows the principles of asthma care, using inhaled corticosteroids and beta-2 agonists within anti-doping limits, complemented by warm-up, trigger avoidance and lifestyle measures.

Fifth, misdiagnosis of EILO as asthma causes prolonged diagnostic delay, ineffective and unnecessary treatment, doping and therapeutic-use-exemption complications, and a substantial psychosocial burden; low awareness among coaches and clinicians is a principal driver of this failure, and education and interdisciplinary communication are practical remedies.

Sixth, the field requires randomised controlled trials of EILO treatments with validated outcome measures, wider access to objective laryngeal diagnosis, and a whole-airway, whole-body, multidisciplinary approach to the breathless athlete. Until these needs are met, the single most important step the clinician can take is to consider the larynx early, to test objectively, and never to treat persistent exertional dyspnoea as asthma without confirming the diagnosis.

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